1 Spectral Graph-of-Graphs Method (specGoG)

Given K related datasets $\{\mathbf{X}^k\}_{k=1}^K$, where $\mathbf{X}^k \in \mathbb{R}^{n\times d_k}$, our goal is to jointly learn the K view Laplacians $\{\mathbf{L}^k\}_{k=1}^K$, their community structures, embedded by $\mathbf{U}^k \in \mathbb{R}^{n\times c_k}$, and a GoG with Laplacian $\mathbf{L} \in \mathbb{R}^{K\times K}$ encoding the relationship between the views. We assume that (i) each $\mathbf{X}^k$ is smooth with respect to $\mathbf{L}^k$; (ii) each view graph can be decomposed as $\mathbf{L}^k = \mathbf{Z}^k + \mathbf{E}^k$, where $\mathbf{Z}^k$ is the low-rank part of the graph that corresponds to the community structure and $\mathbf{E}^k$ is the sparse residual connectivity; and (iii) views with similar spectral embeddings $\mathbf{U}^k$ are strongly connected in the GoG. These assumptions lead to the following spectral graph-of-graphs learning formulation:

$$\begin{aligned}
\min_{\substack{\mathbf{L}^k,\mathbf{Z}^k,\mathbf{E}^k,\\ \mathbf{U}^k,\mathbf{L}}} \quad & \sum_{k=1}^K \operatorname{tr}\Big({\mathbf{X}^k}^\top \mathbf{L}^k \mathbf{X}^k\Big) + \lambda_1 \|\mathcal{Z}\|_{\circledast} + \sum_{k=1}^K \Big[\lambda_2 \|\mathbf{E}^k\|_F^2 + \lambda_3 \operatorname{tr}\Big({\mathbf{U}^k}^\top \mathbf{Z}^k \mathbf{U}^k\Big)\Big] \\
& + \lambda_4 \operatorname{tr}\big(\mathcal{V}(\{\mathbf{U}^k\}_{k=1}^K)^\top \mathbf{L} \mathcal{V}(\{\mathbf{U}^k\}_{k=1}^K)\big) + \lambda_5 \|\mathbf{L}\|_{F,\text{off}}^2 \\
\text{s.t.} \quad & \{\mathbf{L}^k, \mathbf{Z}^k, \mathbf{L}\} \in \mathbb{L}, \quad \operatorname{tr}\left(\mathbf{L}^k\right) = 2n, \quad \operatorname{tr}\left(\mathbf{L}\right) = 2K, \mathbf{L}^k = \mathbf{Z}^k + \mathbf{E}^k, \\
& \mathcal{Z} = \Phi(\mathbf{Z}^1, \ldots, \mathbf{Z}^K), \quad {\mathbf{U}^k}^\top \mathbf{U}^k = \mathbf{I},
\end{aligned} \tag{1}$$

where the first term enforces smoothness of each view's signals over its learned graph, the second term promotes low-rank recovery of the tensor $\mathcal{Z} \in \mathbb{R}^{n\times n\times K}$ via the t-SVD-TNN norm, where $\mathcal{Z} = \Phi(\{\mathbf{Z}^k\}_{k=1}^K)$ stacks the low-rank components of $\mathbf{L}^k$ into a 3-mode tensor and rotates it to $n \times K \times n$ to reduce the cost of updating $\mathcal{Z}$ [**?**]. The third term promotes sparsity of the outlier components, $\mathbf{E}^k$ and the fourth term learns a low-dimensional embedding $\mathbf{U}^k$ of each view from $\mathbf{Z}^k$ with c_k communities. The fifth term couples these embeddings through the GoG Laplacian $\mathbf{L}$, regularizing the stacked embeddings $(\mathcal{V}(\{\mathbf{U}^k\}_{k=1}^K)$ so that structurally similar views are encoded by a large edge weight in $\mathbf{L}$. The last term controls the sparsity of the GoG. The constraints ensure $\mathbf{L}^k$, $\mathbf{Z}^k$, and $\mathbf{L}$ are valid Laplacians with trace normalization fixing the total edge weight, and ${\mathbf{U}^k}^\top \mathbf{U}^k = \mathbf{I}$ enforces orthonormal per-view embeddings. In order to rewrite this objective function with the constraints incorporated into (1), we define the set $\mathcal{H}_1 = \left\{\mathbf{v} \in \mathbb{R}^{n(n-1)/2} \,\middle|\, \mathbf{1}^\top \mathbf{v} = -n\right\}$ and let $\imath_1(\cdot)$ be the indicator function whose value is 0 if its input is in $\mathcal{H}_1$, and ∞ otherwise. Similarly, define the set $\mathcal{H}_2 = \left\{\mathbf{v} \in \mathbb{R}^{n(n-1)/2},\ \mathbf{v} \le 0\right\}$, and its indicator $\imath_2(\cdot)$. Introducing the auxiliary variables $\mathbf{C}^k = \mathbf{L}^k$, $\mathbf{F}^k = \mathbf{Z}^k$, $\mathbf{G} = \mathbf{L}$, and $\mathbf{H} = (\mathcal{V}(\{\mathbf{U}^k\}_{k=1}^K)$, the augmented Lagrangian of the objective function is given by:

$$\begin{aligned}
\min_{\substack{\mathbf{L}^k,\mathbf{Z}^k,\mathbf{E}^k,\mathbf{U}^k,\\ \mathbf{C}^k,\mathbf{F}^k,\mathbf{L},\mathbf{G},\mathbf{H}}} \quad & \sum_{k=1}^K \operatorname{tr}\Big({\mathbf{X}^k}^\top \mathbf{L}^k \mathbf{X}^k\Big) + \lambda_1 \|\mathcal{Z}\|_{\circledast} + \lambda_2 \sum_{k=1}^K \|\mathbf{E}^k\|_F^2 + \lambda_3 \operatorname{tr}\Big({\mathbf{U}^k}^\top \mathbf{F}^k \mathbf{U}^k\Big) + \lambda_4 \operatorname{tr}\big(\mathbf{H}^\top \mathbf{L} \mathbf{H}\big) \\
& + \lambda_5 \|\mathbf{L}\|_{F,\text{off}}^2 + \sum_{k=1}^K \big[\imath_1(\text{up}(\mathbf{L}^k)) + \imath_2(\text{up}(\mathbf{C}^k))\big] + \sum_{k=1}^K \big[\imath_1(\text{up}(\mathbf{Z}^k)) + \imath_2(\text{up}(\mathbf{F}^k))\big] + \imath_1(\text{up}(\mathbf{L})) \\
& + \imath_2(\text{up}(\mathbf{G})) + \sum_{k=1}^K \frac{\alpha}{2} \|\mathbf{L}^k - \mathbf{Z}^k - \mathbf{E}^k + \frac{\mathbf{Y}_1^k}{\alpha}\|_F^2 + \sum_{k=1}^K \left[\frac{\alpha}{2} \|\mathbf{C}^k - \mathbf{L}^k + \frac{\mathbf{Y}_2^k}{\alpha}\|_F^2 + \frac{\alpha}{2} \|\mathbf{F}^k - \mathbf{Z}^k + \frac{\mathbf{Y}_3^k}{\alpha}\|_F^2\right] \\
& + \frac{\alpha}{2} \|\mathbf{G} - \mathbf{L} + \frac{\mathbf{Y}_4}{\alpha}\|_F^2 + \frac{\alpha}{2} \|\mathbf{H} - (\mathcal{V}(\{\mathbf{U}^k\}_{k=1}^K) + \frac{\mathbf{Y}_5}{\alpha}\|_F^2, \\
\text{s.t.} \quad & {\mathbf{U}^k}^\top \mathbf{U}^k = \mathbf{I},
\end{aligned} \tag{2}$$

where $\{\mathbf{Y}_1^k, \mathbf{Y}_2^k, \mathbf{Y}_3^k, \mathbf{Y}_4, \mathbf{Y}_5\}$ are Lagrange multipliers, and α is the penalty parameter. The proposed objective function can be solved using the ADMM. The solution of the proposed method as follows:

- $\mathbf{L}^k$ **update:** The $\mathbf{L}^k$ subproblem can be formulated as follows:

$$\min_{\mathbf{L}^k} \quad \sum_{k=1}^K \operatorname{tr}\Big({\mathbf{X}^k}^\top \mathbf{L}^k \mathbf{X}^k\Big) + \imath_1(\text{up}(\mathbf{L}^k)) + \frac{\alpha}{2} \|\mathbf{L}^k - \mathbf{Z}^k - \mathbf{E}^k + \frac{\mathbf{Y}_1^k}{\alpha}\|_F^2 + \frac{\alpha}{2} \|\mathbf{C}^k - \mathbf{L}^k + \frac{\mathbf{Y}_2^k}{\alpha}\|_F^2, \tag{3}$$

Let $\mathbf{b}^k = \text{up}(\mathbf{X}^k {\mathbf{X}^k}^\top)$, $\mathbf{d}^k = \text{diag}(\mathbf{X}^k {\mathbf{X}^k}^\top)$, $\ell^k = \text{up}(\mathbf{L}^k)$, $\mathbf{z}^k = \text{up}(\mathbf{Z}^k)$, $\mathbf{e}^k = \text{up}(\mathbf{E}^k)$, $\mathbf{y}_1^k = \text{up}(\mathbf{Y}_1^k)$, $\mathbf{c}^k = \text{up}(\mathbf{C}^k)$, $\mathbf{y}_2^k = \text{up}(\mathbf{Y}_2^k)$. Then, reformulate (3) as follows:

$$\begin{aligned}
\min_{\ell^k} \quad & \sum_{k=1}^K (2\mathbf{b}^k - \mathbf{S}^\top \mathbf{d}^k)^\top \ell^k + \frac{\alpha}{2} \|\ell^k - \mathbf{z}^k - \mathbf{e}^k + \frac{\mathbf{y}_1^k}{\alpha}\|_F^2 + \frac{\alpha}{2} \|\mathbf{c}^k - \ell^k + \frac{\mathbf{y}_2^k}{\alpha}\|_F^2, \\
\text{s.t.} \quad & \mathbf{1}^\top \ell^k = -n.
\end{aligned} \tag{4}$$

where $\mathbf{S} \in \mathbb{R}^{n \times n(n-1)/2}$ is the row-sum operator that maps the strictly upper triangular entries of a symmetric matrix to its row sums, excluding the diagonal entries. The solution can be found as follows:

$$\hat{\ell}^k = \Pi_1\Big(\frac{\mathbf{S}^\top \mathbf{d}^k - 2\mathbf{b}^k + \alpha \mathbf{z}^k + \alpha \mathbf{e}^k - \mathbf{y}_1^k + \alpha \mathbf{c}^k + \mathbf{y}_2^k}{2\alpha}\Big). \tag{5}$$

where $\Pi_1(\cdot)$ is the projection onto $\mathcal{H}_1$.

- $\mathbf{E}^k$ **subproblem:** The $\mathbf{E}^k$ subproblem can be formulated as follows:

$$\min_{\mathbf{E}^k} \quad \lambda_2 \|\mathbf{E}^k\|_F^2 + \frac{\alpha}{2}\|\hat{\mathbf{L}}^k - \mathbf{Z}^k - \mathbf{E}^k + \frac{\mathbf{Y}_1^k}{\alpha}\|_F^2. \tag{6}$$

Equivalently,

$$\min_{\mathbf{e}^k} \quad \lambda_2 {\mathbf{e}^k}^\top (\mathbf{S}^\top \mathbf{S} + 2\mathbf{I})\mathbf{e}^k + \frac{\alpha}{2}\|\hat{\ell}^k - \mathbf{z}^k - \mathbf{e}^k + \frac{\mathbf{y}_1^k}{\alpha}\|_F^2. \tag{7}$$

The solution can be written as follows:

$$\hat{\mathbf{e}}^k = (2\lambda_2 \mathbf{S}^\top \mathbf{S} + (4\lambda_2 + \alpha)\mathbf{I})^{-1}(\alpha \hat{\ell}^k - \alpha \mathbf{z}^k + \mathbf{y}_1^k). \tag{8}$$

- $\mathbf{Z}^k$ **subproblem:** The $\mathbf{Z}^k$ subproblem can be formulated as follows:

$$\begin{aligned} \min_{\mathbf{Z}^k} \quad & \lambda_1 \|\mathcal{Z}\|_{\circledast} + \sum_{k=1}^{K} \left[\imath_1(\mathrm{up}(\mathbf{Z}^k)) + \frac{\alpha}{2}\|\hat{\mathbf{L}}^k - \mathbf{Z}^k - \hat{\mathbf{E}}^k + \frac{\mathbf{Y}_1^k}{\alpha}\|_F^2 + \frac{\alpha}{2}\|\mathbf{F}^k - \mathbf{Z}^k + \frac{\mathbf{Y}_3^k}{\alpha}\right] \\ \text{s.t.} \quad & \mathcal{Z} = \Phi(\mathbf{Z}^1, \ldots, \mathbf{Z}^K), \qquad \mathbf{Z}^k = {\mathbf{Z}^k}^\top, \end{aligned} \tag{9}$$

Equivalently,

$$\begin{aligned} \min_{\mathbf{Z}^k} \quad & \lambda_1 \|\mathcal{Z}\|_{\circledast} + \sum_{k=1}^{K} \left[\imath_1(\mathrm{up}(\mathbf{Z}^k)) + \frac{\alpha}{2}\|\mathbf{Z}^k - \mathbf{R}^k\|_F^2 \right] \\ \text{s.t.} \quad & \mathcal{Z} = \Phi(\mathbf{Z}^1, \ldots, \mathbf{Z}^K), \qquad \mathbf{Z}^k = {\mathbf{Z}^k}^\top, \end{aligned} \tag{10}$$

where $\mathbf{R}^k = \frac{1}{2}(\hat{\mathbf{L}}^k - \hat{\mathbf{E}}^k + \mathbf{F}^k + \frac{\mathbf{Y}_1^k + \mathbf{Y}_3^k}{\alpha})$. The solution $\mathbf{Z}^k$ can be found by applying tensor tubal-shrinkage operator [**?**] on tensor $\mathcal{R} = \Phi(\mathbf{R}^1, \ldots, \mathbf{R}^k)$, and then project it to $\imath_1(\cdot)$.

- $\mathbf{F}^k$ **subproblem:** The $\mathbf{F}^k$ subproblem can be formulated as follows:

$$\min_{\mathbf{F}^k} \quad \sum_{k=1}^{K} \lambda_3 \,\mathrm{tr}\Big({\mathbf{U}^k}^\top \mathbf{F}^k \mathbf{U}^k\Big) + \imath_2(\mathrm{up}(\mathbf{F}^k)) + \frac{\alpha}{2}\|\mathbf{F}^k - \hat{\mathbf{Z}}^k + \frac{\mathbf{Y}_3^k}{\alpha}\|_F^2. \tag{11}$$

Let $\mathbf{b}_1^k = \mathrm{up}(\mathbf{U}^k {\mathbf{U}^k}^\top)$, $\mathbf{d}_1^k = \mathrm{diag}(\mathbf{U}^k {\mathbf{U}^k}^\top)$, and $\mathbf{y}_3^k = \mathrm{up}(\mathbf{Y}_\mathbf{3}^k)$. Then, the $\mathbf{F}^k$ can be reformulated as follows:

$$\min_{\mathbf{f}^k} \quad \sum_{k=1}^{K} \lambda_3 (2\mathbf{b}_1^k - \mathbf{S}^\top \mathbf{d}_1^k)^\top \mathbf{f}^k + \frac{\alpha}{2}\|\mathbf{f}^k - \hat{\mathbf{z}}^k + \frac{\mathbf{y}_3^k}{\alpha}\|_F^2 + \imath_2(\mathrm{up}(\mathbf{F}^k)), \tag{12}$$

The solution can be found as follows:

$$\hat{\mathbf{f}}^k = \Pi_2\Big(\frac{\lambda_3(\mathbf{S}^\top \mathbf{d}_1^k - 2\mathbf{b}_1^k) + \alpha \hat{\mathbf{z}}^k - \mathbf{y}_3^k}{\alpha}\Big), \tag{13}$$

where $\Pi_2(\cdot)$ is the projection onto $\mathcal{H}_2$.

- $\mathbf{C}^k$ **subproblem:** The $\mathbf{C}^k$ subproblem can be formulated as follows:

$$\min_{\mathbf{C}^k} \quad \sum_{k=1}^{K} \imath_2(\mathrm{up}(\mathbf{C}^k)) + \frac{\alpha}{2}\|\mathbf{C}^k - \hat{\mathbf{L}}^k + \frac{\mathbf{Y}_2^k}{\alpha}\|_F^2. \tag{14}$$

(14) can be written in vectorized form:

$$\hat{\mathbf{c}}^k = \Pi_2\Big(\hat{\ell}^k - \frac{\mathbf{y}_2^k}{\alpha}\Big), \tag{15}$$

- $\mathbf{L}$ **subproblem:** The $\mathbf{L}$ subproblem can be formulated as follows:

$$\min_{\mathbf{L}} \quad \lambda_4 \operatorname{tr}\big(\mathbf{H}^\top \mathbf{L}\mathbf{H}\big) + \lambda_5 \|\mathbf{L}\|_{F,\text{off}}^2 + \frac{\alpha}{2}\|\mathbf{G} - \mathbf{L} + \frac{\mathbf{Y}_4}{\alpha}\|_F^2 + \imath_1(\mathrm{up}(\mathbf{L})). \tag{16}$$

Let $\mathbf{b} = \mathrm{up}(\mathbf{H}\mathbf{H}^\top)$, $\mathbf{d} = \mathrm{diag}(\mathbf{H}\mathbf{H}^\top)$, and $\mathbf{y}_4 = \mathrm{up}(\mathbf{Y}_4^{\mathbf{k}})$. Then, the $\mathbf{L}$ can be reformulated as follows:

$$\min_{\ell} \quad \lambda_4(2\mathbf{b} - \mathbf{S}_1^\top \mathbf{d})^\top \ell + \lambda_5 \ell^\top (\mathbf{S}_1^\top \mathbf{S}_1 + 2\mathbf{I})\ell + \frac{\alpha}{2}\|\mathbf{g} - \ell + \frac{\mathbf{y}_4}{\alpha}\|_F^2 + \imath_1(\mathrm{up}(\ell)), \tag{17}$$

where $\mathbf{S}_1 \in \mathbb{R}^{K \times K(K-1)/2}$ is the row sum operator that maps the strictly upper triangular entries of a symmetric matrix to its row sums, excluding the diagonal entries. The solution of (17) can be found as follows:

$$\hat{\ell} = \Pi_1\big((2\lambda_5 \mathbf{S}_1^\top \mathbf{S}_1 + [4\lambda_5 + \alpha]\mathbf{I}_{\frac{K(K-1)}{2}})^{-1}(\lambda_4(\mathbf{S}_1^\top \mathbf{d} - 2\mathbf{b}) + \alpha\mathbf{g} + \mathbf{y}_4)\big). \tag{18}$$

- $\mathbf{G}$ **subproblem:** The $\mathbf{G}$ subproblem can be formulated as follows:

$$\min_{\mathbf{G}} \quad \imath_2(\mathrm{up}(\mathbf{G})) + \frac{\alpha}{2}\|\mathbf{G} - \hat{\mathbf{L}} + \frac{\mathbf{Y}_4}{\alpha}\|_F^2. \tag{19}$$

(19) can be written in vectorized form as follows:

$$\min_{\mathbf{g}} \quad \imath_2(\mathrm{up}(\mathbf{G})) + \frac{\alpha}{2}\|\mathbf{g} - \hat{\ell} + \frac{\mathbf{y}_4}{\alpha}\|_F^2. \tag{20}$$

The solution of (20) can be found as follows:

$$\hat{\mathbf{g}} = \Pi_2\Big(\hat{\ell} - \frac{y_4}{\alpha}\Big). \tag{21}$$

- $\mathbf{H}$ **subproblem:** The $\mathbf{H}$ subproblem can be formulated as follows:

$$\min_{\mathbf{H}} \quad \lambda_4 \operatorname{tr}\Big(\mathbf{H}^\top \hat{\mathbf{L}}\mathbf{H}\Big) + \frac{\alpha}{2}\|\mathbf{H} - (\mathcal{V}(\{\mathbf{U}^k\}_{k=1}^K) + \frac{\mathbf{Y}_5}{\alpha}\|_F^2. \tag{22}$$

The solution of (22) can be found as follows:

$$\hat{\mathbf{H}} = (2\lambda_4 \hat{\mathbf{L}} + \alpha \mathbf{I}_K)^{-1}\big(\alpha \mathcal{V}(\{\mathbf{U}^k\}_{k=1}^K) - \mathbf{Y}_5\big). \tag{23}$$

- $\mathbf{U}^k$ **subproblem:** The $\mathbf{U}^k$ subproblem can be formulated as follows:

$$\begin{aligned} \min_{\mathbf{U}^k} \quad & \sum_{k=1}^{K} \lambda_3 \operatorname{tr}\left(\mathbf{U}^{k^\top} \hat{\mathbf{F}}^k \mathbf{U}^k\right) + \frac{\alpha}{2} \|\hat{\mathbf{H}} - (\mathcal{V}(\{\mathbf{U}^k\}_{k=1}^K) + \frac{\mathbf{Y}_5}{\alpha}\|_F^2, \\ \text{s.t.} \quad & \mathbf{U}^{k^\top} \mathbf{U}^k = \mathbf{I}, \end{aligned} \tag{24}$$

Equivalently,

$$\begin{aligned} \min_{\mathbf{U}^k} \quad & \sum_{k=1}^{K} \lambda_3 \operatorname{tr}\left(\mathbf{U}^{k^\top} \hat{\mathbf{F}}^k \mathbf{U}^k\right) + \frac{\alpha}{2} \|\hat{\mathbf{H}}^k - \mathbf{U}^k + \frac{\mathbf{Y}_5^k}{\alpha}\|_F^2, \\ \text{s.t.} \quad & \mathbf{U}^{k^\top} \mathbf{U}^k = \mathbf{I}, \end{aligned} \tag{25}$$

The $\mathbf{U}^k$ subproblem in (25) is solved using generalized power iteration (GPI). By expanding the Frobenius norm and removing constant terms, the problem can be rewritten as

$$\begin{aligned} \min_{\mathbf{U}^k} \quad & \operatorname{tr}\left(\mathbf{U}^{k^\top} \mathbf{\Gamma}^k \mathbf{U}^k\right) - 2 \operatorname{tr}\left(\mathbf{U}^{k^\top} \mathbf{\Psi}^k\right), \\ \text{s.t.} \quad & \mathbf{U}^{k^\top} \mathbf{U}^k = \mathbf{I}, \end{aligned} \tag{26}$$

where $\mathbf{\Gamma}^k = \lambda_3 \hat{\mathbf{F}}^k$ and $\mathbf{\Psi}^k = \frac{\alpha}{2} \hat{\mathbf{H}}^k + \frac{1}{2} \mathbf{Y}_5^k$.
To apply GPI, we define $\mathbf{M}^k = \gamma^k \mathbf{I} - \mathbf{\Gamma}^k$, where $\gamma^k > \lambda_{\max}(\mathbf{\Gamma}^k)$. At each iteration, we compute

$$\mathbf{Z}^{k^{(t)}} = \mathbf{M}^k \mathbf{U}^{k^{(t)}} + \mathbf{\Psi}^k. \tag{27}$$

Applying the singular value decomposition (SVD), $\mathbf{Z}^{k^{(t)}} = \mathbf{P}^k \mathbf{\Sigma}^k \mathbf{Q}^{k^\top}$, the update is given by

$$\hat{\mathbf{U}}^{k^{(t+1)}} = \mathbf{P}^k \mathbf{Q}^{k^\top}. \tag{28}$$

The iterations continue until convergence, while preserving the orthogonality constraint $\mathbf{U}^{k^\top} \mathbf{U}^k = \mathbf{I}$.

- **Lagrangian multiplier and penality parameter**: The Lagrangian multiplier and penality parameter can be updated as follows:

$$\begin{aligned} \hat{\mathbf{Y}}_1^k &= \mathbf{Y}_1^k + \alpha \left(\hat{\mathbf{L}}^k - \hat{\mathbf{Z}}^k - \hat{\mathbf{E}}^k\right), \\ \hat{\mathbf{Y}}_2^k &= \mathbf{Y}_2^k + \alpha \left(\hat{\mathbf{C}}^k - \hat{\mathbf{L}}^k\right), \\ \hat{\mathbf{Y}}_3^k &= \mathbf{Y}_3^k + \alpha \left(\hat{\mathbf{F}}^k - \hat{\mathbf{Z}}^k\right), \\ \hat{\mathbf{Y}}_4 &= \mathbf{Y}_4 + \alpha \left(\hat{\mathbf{G}} - \hat{\mathbf{L}}\right), \\ \hat{\mathbf{Y}}_5 &= \mathbf{Y}_5 + \alpha \left(\hat{\mathbf{H}} - \mathcal{V}(\{\hat{\mathbf{U}}^k\}_{k=1}^K)\right). \end{aligned} \tag{29}$$

The penalty parameter is updated as

$$\hat{\alpha} = \rho \alpha, \quad \rho > 1. \tag{30}$$